

Si Jine Roh

SOFTWARE ENGINEER

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Senior Backend Software Engineer with over a decade of experience designing and scaling high-performance C#/.NET APIs, platform services, and shared libraries. Specializes in real-time industrial software for autonomous drilling and robotics systems. Fully owns backend architecture for mission-critical SLB applications, driving API contract design, release readiness, and production support to ensure flawless live well-construction operations.

Career

Schlumberger

SENIOR SOFTWARE ENGINEER

Houston, Texas, US

Nov 2024 - Present

- Lead C#/.NET REST API and shared-library design for autonomous drilling guidance and drilling operations clients
- Own service contracts, API versioning, and backward compatibility for release-critical live drilling workflows
- Support two major releases per year, 1-3 minor releases or patches per major release, and monthly remote-operation service releases
- Resolve QA and field failures through defect reproduction, backend fixes, and release validation improvements

Schlumberger

SOFTWARE ENGINEER

Houston, Texas, US

Aug 2022 - Nov 2024

- Shipped C#/.NET REST endpoints and shared libraries for autonomous drilling advisor and drilling operations workflows
- Built backend features for deviation performance, anti-collision reporting, drilling plan validation, and boundary synchronization
- Expanded backend validation by adding and updating end-to-end, anti-collision, simulator, and CI pipeline test cases
- Investigated field-support issues across APIs, persisted data, validation logic, message queues, and client integration paths
- Supported deliverable-generation services, major releases, patch releases, and post-deployment support escalations

Schlumberger

SOFTWARE ENGINEER

Beijing, China

Jun 2019 - Aug 2022

- Designed REST/GraphQL APIs and shared libraries consumed by 6+ product areas across drilling operations workflows
- Built Azure Storage and MongoDB-backed catalog/data workflows with pipeline validation, rollback behavior, and deployment support
- Updated GraphQL APIs and shared platform libraries to improve service security, compatibility, and maintainability
- Coordinated weekly production releases, design reviews, planning, and support handoffs between Beijing and U.S. teams

Schlumberger

SOFTWARE ENGINEER

Katy, Texas, US

Feb 2018 - Jun 2019

- Built C# drilling-data analysis modules for inferred build-rate and turn-rate behavior in early autonomous workflows
- Implemented unit-tested service logic for physics-based drilling tendency prediction and field-support diagnostics
- Investigated Tier-2 backend issues, connecting field behavior to service logic and data-quality failures
- Contributed software concepts later reflected in PCT-published autonomous wellbore deviation detection and handling

ColubrisMX, Inc.

SOFTWARE ENGINEER

Houston, Texas, US

Mar 2017 - Feb 2018

- Engineered real-time motor-control software in C++ and TwinCAT for a surgical robotic-arm for neurovascular surgical procedures
- Built a robot-arm testing tool that reduced tuning time by 50%
- Established build, issue-tracking, and version-control infrastructure from scratch for a small engineering team
- Operated and cooperated multiple surgical robot testing scenarios and demonstrations

Department of Neurosurgery, University of Texas Health Science Center at Houston

RESEARCH ASSISTANT II

Houston, Texas, US

Jun 2016 - Feb 2017

- Engineered real-time motor-control software in C++ and TwinCAT for a surgical robotic-arm for neurovascular surgical procedures
- Operated and cooperated multiple surgical robot testing scenarios and demonstrations

Department of Computer Science & Engineering, Texas A&M University

UNDERGRADUATE RESEARCHER

- Designed and developed UI for UI/UX research to collect user feedback based on UI/UX differences

College Station, Texas, US

Jan 2016 - May 2016

PEER TEACHER

- Coached students to debug and design program during lab sessions and office hours
- Received Certificate of Appreciation

Sep 2015 - May 2016

Korean Students Association, Texas A&M University

WEB DEVELOPER & ADMINISTRATOR

- Designed and developed Korean Students Association website

College Station, Texas, US

May 2014 - May 2016

O'square Co., Ltd.

WEB & ANDROID DEVELOPER

- Developed social activity recommend system in Ruby and Rails on AWS
- Developed Android application with native development toolkit

Seoul, Korea

Aug 2012 - Mar 2013

Patents and Publications

Wellbore Drill Deviation Handling

PATENT

Published Mar 6, 2025

- Co-inventor on a pending patent publication covering detection, alerting, and automated handling of out-of-tolerance drill deviation during wellbore drilling
- US Patent US12590526B2; related PCT\WO (WO2025049755A1) and China (CN121889566A) applications pending

AI Solution Provides the Most Autonomous Framework of Directional Well Sections in High-DLS Well Plans

SPE/IADC DRILLING CONFERENCE AND EXHIBITION (PAGES D011S001R003)

Published Feb 25, 2025

- Co-author; contributed software architecture and backend integration context for autonomous control workflows

Advancement in Autonomous Directional Drilling for Curve and Tangent

SPE/IADC DRILLING CONFERENCE AND EXHIBITION (PAGES D012S001R004)

Published Feb 27, 2024

- Co-author; contributed autonomous drilling software and backend workflow context

Education

Texas A&M University

B.S. IN COMPUTER ENGINEERING

College Station, Texas, US

May 2016

- Graduated with Cum Laude (Cumulative GPA: 3.55/4.0)
- Certificate of Appreciation, Department of Computer Science & Engineering, Texas A&M University, Spring 2016
- Dean's Honor Award, The Dwight Look College of Engineering, Texas A&M University, Spring 2015